Introduction to systems administration

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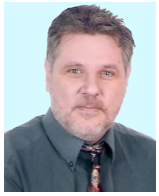
Universitat Oberta de Catalunya



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Introduction

This module provides an overview of how IT systems are structured within an organisation. It also describes the physical components that constitute these systems, as well as the human component. These elements combined form what organisations refer to as their ***IT department***.

This module describes why and how the learning resources have been divided into different modules, and their connection to the organisation's functions and organisational chart. It is worth noting that **IT systems** are just another cog in the organisational wheel, although they have gradually become a **cornerstone** in every department of an organisation.

Objectives

The learning resources in this module cover the contents needed to achieve the following objectives:

1. To be familiar with the key members of the IT department.

2. To have a clear understanding of the various roles that may exist within an IT department.

3. To learn how different organisational structures are possible, although they will be organised to carry out the functions corresponding to the IT department.

4. To learn that roles can be divided into one or more jobs, according to the organisation's needs.

5. To understand the IT manager's role within the department and organisation.

1. IT systems and the organisation

We can analyse the organisation's **IT department** from various perspectives:

- **Organisational**: a department like any other, with resources available to the organisation.
- **Technical**: a collection of servers, networks and workstations that enable the organisation's users (employees) to do their jobs.
- **User**: an additional tool provided by the organisation to improve their work.
- **Management**: a large query database that aids decision-making.

All perspectives matter when it comes to building an effective and efficient IT system. A system is considered effective when it produces the expected results and efficient when it performs the assigned task within a reasonable time frame using an appropriate number of resources.

However, the organisational level is important to consider in our case, as this is where **IT managers** are positioned. As the following modules will show, IT managers are ultimately responsible for the IT department and represent the final decision-makers regarding the organisation's IT system.

IT Department

The term *information technology* (IT) is currently used to refer to all aspects of computing. For simplicity's sake, the IT department will be treated as a separate entity.

Positions

The person in charge of the IT department is usually called the *IT director*, *Head of IT* or ***IT manager***. From here on, we will use the term *IT manager* to refer to this position.

Some organisations use the term *information technology* (IT), in which case there will probably be an IT management department and an IT director or IT manager. IT encompasses more than just computing.

An IT department's key task is to plan and manage the IT system to ensure it provides good service to users, as covered in the *user administration* module and other learning resources.

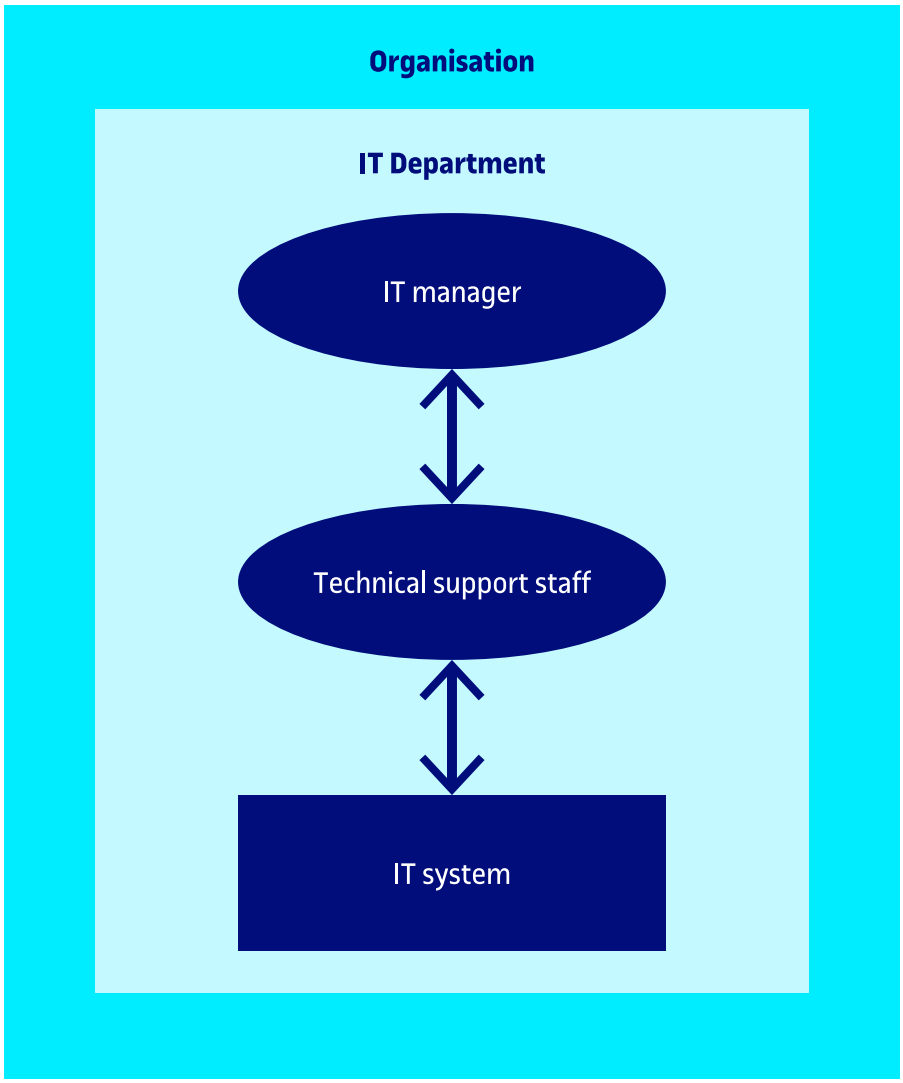
See also

See the *The IT system within the organisation* module for more information about their role.

2. IT systems: elements

As shown in Figure 1, the IT department is composed of three core elements: the IT manager, the technical support staff and the IT system.

Figure 1. IT department organisational chart



Below there is a description of the different elements that composes the IT department.

1) IT manager. As head of the IT department, they are responsible for making organisational decisions that affect the IT system, as well as managing technical support staff.

2) Technical support staff. Responsible for system maintenance and user support.

3) Physical IT system. It is made up of two main components: hardware and software:

a) Hardware: this is all the computer equipment necessary to make the IT system work properly. Broadly speaking, we can break down hardware into three components:

- **Servers**. Remote computers integrated into the organisation's network that contain programs and data, which users access remotely through the organisation's network. Multiple servers are common in organisations because they specialise in different functions. This increases control by system administrators and reduces the risk of failure.

- **Workstations**. A workstation is the computer that the user (worker) is assigned to within the organisation. Nowadays each user is usually assigned a workstation, which means that there is a very high volume of workstations installed.

- **Network cabling and electronics**. **Network** or **network cabling** refers to both the physical cable connecting the workstations and the network control electronics, made up of hubs, routers, firewalls and switches.

See also

Network cabling and network electronics are covered in the module *Network administration*, although a full and detailed explanation is given in the networking subjects.

b) Software: this refers to all the programs and applications that the user will need to work. Generally, it can be broken down into four categories:

- **Basic software**. This includes the operating system and the basic communications applications on the servers.

- **Office software**. These are the office suites, which usually include a spreadsheet, a word processor, a database and an agenda.

- **Communications software**. These usually include e-mail, an internet browser (Edge, Firefox, Chrome, etc.) and software for making video calls (Teams, Hangouts, etc.).

- **Specific applications**. This group of applications includes programs that are organisation- or even department-dependent. Invoicing, accounting, graphic design, production control, payroll programs, etc.

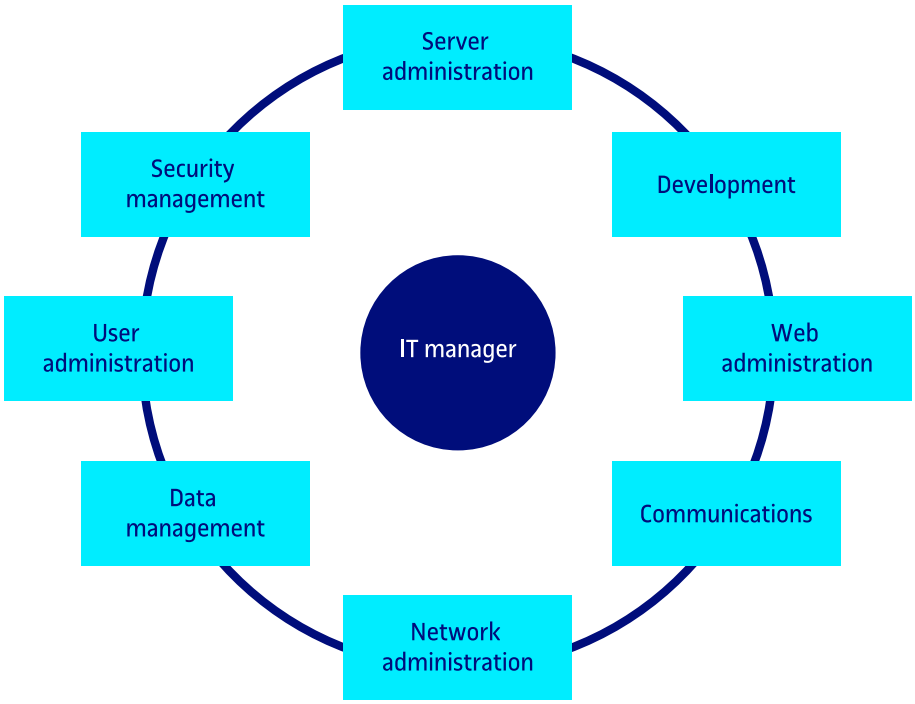
In the following modules, these concepts will be explored and developed in greater detail, and their relationship with each other will be discussed, since, for an IT system to function correctly, all the components that make it up must be closely linked.

3. IT system staff

An organisational chart of the department could be drawn up, but organisations are conditioned by their respective cultures and practices. Therefore, discussing the tasks to be carried out in the department is much more useful, after the organisation itself distributes them within its own organisational chart.

Although organisations differ greatly from one another, all tasks to be performed by an IT department are commonly grouped into related groups of tasks, which form the building blocks of these resources. Figure 2 shows the different tasks found in the IT department.

Figure 2. IT department tasks in an organisation



All IT department tasks are function-specific. Therefore, they are all covered by one or two people in many organisations, and only large organisations have several people (or sub-departments) dedicated to each of these functions. Some organisations may not have any of these tasks. For example, many organisations outsource software development to other companies (consultancies), so this task will not be necessary.

3.1. Roles

Below is a brief description of what each of these roles in the IT department entails:

1) Server administration. Server administrators are tasked with installing, maintaining and servicing the organisation's servers and IT infrastructure as a whole. They must also make sure that the servers can be recovered quickly in the event of a failure. In addition, they must keep the servers up and running so that all other administration tasks (e.g. user administration) can be carried out.

2) User administration. This role is all about taking care of users' needs and maintaining their desktops. Since the IT system is still an organisational tool, this is one of the most visible parts of the system. Therefore, it is important to take great care because it is easy for the user to experience issues with the day-to-day operation of the system.

3) Network administration. The core part of this role concerns the physical part of the network, ensuring that it is working properly and monitoring that it is in good condition, as well as focusing on upgrading it and delivering it to all areas of the organisation.

4) Data management. This role is tasked with maintaining the integrity of the organisation's information. It may be located on servers or scattered throughout the IT system. It is often a mixture of both, i.e. many components are on the servers, but there is often important information scattered on workstations. When large databases exist, protecting the integrity of the information also makes sense. Also, linking the stored information is very useful in order to know more about the organisation.

5) Web administration. This role includes the maintenance of the web server, and in many cases also of the content (of the webpages). Addtionally, this role maintains the overall consistency across the server.

6) Security management. The role of IT security is complex, ranging from securing existing information to physically protecting equipment from theft, as well as preventing internet-borne attacks on IT systems. As a matter of fact, security is a task present in all roles, as it is a natural part of IT. The organisation's information needs to be protected.

7) Development. The organisation requires specific software and it is important to be careful when developing or acquiring it. This is the role of the development department if only new software can be covered by the manufacturer. Development will not be covered in these learning resources, as there are subjects focused to how software is made and how IT projects are carried out.

The only things to be covered are the relationship between the IT manager and how to implement new software in the organisation, as well as the agile methodologies that can be adopted in the organisation to carry out this task

efficiently. However, keep in mind that there are other solutions suggested by the technical support staff, albeit it is the IT manager who must make the final decision, which are as follows:

- **Purchasing standard software**. There are often standard invoicing programs, or ERP's such as SAP, which have been adopted as *de facto* software in organisations.

- **Outsourcing the application to a company outside the organisation.** Depending on the type of software to be developed, organisations often hire third party companies (consultancies) to develop it. This is more economical in terms of resources than having the organisation's staff dedicated to developing software.

8) IT manager. This is the most senior manager in the IT department. This role, together with server administrator and user administrator, are the most important roles in the department. The IT manager liaises between the needs of the organisation and the work done in the department. They therefore act as the bridge between the two places. Commonly the IT manager, together with (or advised by) the server administrators and user administrators, makes the decisions that affect the organisation's IT structure in the medium and long term (the ultimate responsibility is his or her alone): from what software to buy (or whether and how it has to be customised) or the servers needed to be installed for the users. A thorough review of all their tasks is beyond the scope of these resources, so only some of their aspects will be covered, such as deciding what software to buy, since keeping IT up to date is a key, complex and financially compromising task.

IT manager

This often includes server administrator, user administrator and the umbrella term system administrator.

An **IT department** does not necessarily require every role. The most essential are IT manager, user administrator and server administrator, which can be performed by a single person.

These resources are split into different modules to explain each of these roles in detail, due to the complexity of each of them. Each of these roles will be covered in detail throughout the different modules that make up these materials, and their relationship to each other will be discussed.

3.2. Role overlapping

All these roles have defined tasks, although they do not work independently, but rather in a coordinated manner, and often the roles overlap at some points. Therefore, each organisation defines which role is assigned to which tasks in case of multiple roles.

Does registering a user on the servers fall under the role of the user administrator or the server administrator? If it is the same person who performs both roles, there is no

problem. If it is not, it should be decided who does it. If the person who administers the servers does it, the task of registering users (and all that goes with it) has been divided between several people. It has to be assessed whether it is worth it. If only the person who administers the users does it, then they will "tweak" the servers to do it. You need to assess whether it is a wise approach.

The solution to the concern in the example above, and to many other problems in administration, is that tasks must cross over, and a user administrator must be able to register new users for efficient administration. This means interacting with the servers (which the server administrator does).

This problem is therefore generalised to all areas of IT systems administration. Therefore, although it has been listed separately here, the reality is that some tasks must be combined to achieve the best possible management, service and user support.

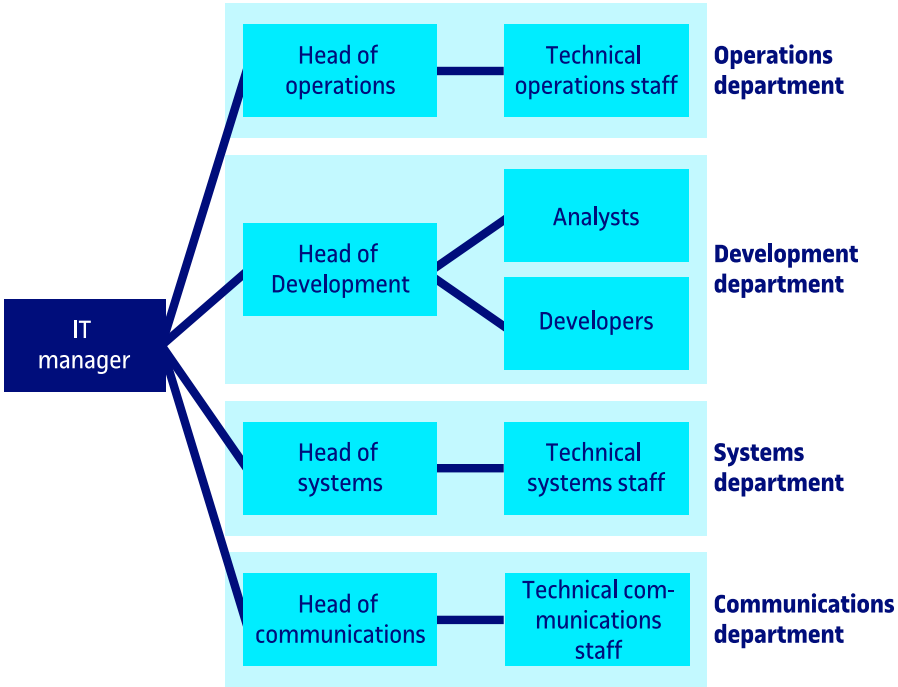
Reflection

How are functions divided up in your organisation? You can leave a comment in the subject forum.

3.3. Organisation chart

Figure 3 shows an example of a potential IT department organisation chart. However, it is the culture, background and habits within the organisation that create the actual organisation chart. Technology is another factor that has a strong influence because it can introduce changes due to shifting needs.

Figure 3. Potential IT department organisational chart



Summary

IT has become an essential part of any organisation, and if this organisation reaches a certain size, its computers cannot be isolated. Therefore, there will be a network and an IT department will be necessary to optimise their use.

Generally , the IT department is composed of three core elements: the IT manager, the technical support staff and the IT system. The IT manager is the senior manager of the IT department and is responsible for managing the IT system at the organisational level.

The size of this department will depend on the organisation's complexity, size and purpose, but the tasks to be performed will always be the same. A large organisation will need many people to carry out any of the roles, while a small organisation may have only one person to cover several roles.

Finally, how the necessary roles are distributed in the actual organisational chart highly depends on the particular organisation.

Activities

1. How could the IT department be arranged in the following organisations?

a) Hospital
b) Car manufacturing company
c) Department store

Self-evaluation

1. How can the organisation's management understand the IT system?

a) As a set of interconnected resources that help it to improve performance.
b) As a large database that helps it to make decisions.
c) As a production system.
d) b and c.

2. The person responsible for ensuring that the workstations have all the software they need is...

a) the IT manager.
b) the server administrator.
c) the network administrator.
d) the user administrator.

3. To whom would you assign the task of maintaining the workstations?

Answer key

Self-evaluation

1. b

2. d

3. User administrator.

Glossary

core software A set of programs that all workstations have as standard. This generally includes the operating system, office packages, browsers, email clients and other standard packages.

database A computerised repository of information in which data can be easily stored, retrieved and linked together.

desktop See **workstation**.

Head of IT See **IT manager**.

Head of the IT department See **IT manager**.

information technology A set of techniques for processing information in whatever format it may be displayed. This also includes computing.
acronym: IT

IT See **information technology**.

IT director See **IT manager**.

IT manager Head of the organisation's IT department.
syn.: Head of the IT department, IT director, Head of IT

microcomputing Name given to the personal computer when the server is a large host (macrocomputer, mainframe). See **workstation**.

organisation Any entity, institution or group that needs or uses an IT infrastructure to carry out its mission.

server A computer dedicated to storing information or applications and serving user requests over a network.

specific applications A set of applications or programs that the organisation has purchased and therefore are not considered standard, as they have been configured or adapted according to the organisation's needs and may not be installed on all workstations or may only be used by certain users in the organisation. They are considered as special software, which generally includes access to organisational databases, payroll, accounting, invoicing programs, etc.

systems administrator Person in charge of IT, who often carries out all tasks, but especially server administrator and user administrator, and sometimes also IT manager.

workstation A set of computers that users operate, which are generally distributed throughout the organisation, typically in offices or production areas. They may be PCs, Macs, Linux, laptops, point-of-sale terminals, etc.

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